SPOTIFY DAILY TRANSACTION ANALYTICS

Subscription & In-App Transaction Demand Analytics — Descriptive Analytics Report

Report Focus Keywords: Top Transaction Types, Revenue Concentration, Volume Trends, Regional Demand Distribution, Channel Tier Allocation, Customer Segment Share, Payment Rail Preferences, Transaction Success Rate, Renewal-Risk Buffer, Descriptive-to-Predictive Transition.

Dataset:	Spotify Daily Transaction Data	**Time Period:**	30 days (2026-07-31 to 2026-08-29)
Records:	60 markets x 10 txn types x 8 regions	**Gross Transaction Value:**	~INR 54.7 Cr
Analytical Context	Streaming Subscription & Commerce Operations	**Analytical Stage:**	Descriptive Analytics Report

Key Performance Indicators (KPI Snapshot)

Headline operating metrics for the 30-day observation window, computed directly from the aggregated transaction dataset.

1.94M	**INR 54.7 Cr**	**INR 282**	**97.3%**
Total Transactions (30 Days)	Gross Transaction Value	Avg. Transaction Value (ATV)	Avg. Transaction Success Rate
18 / 60	**23.4%**	**13.8%**	**32.3%**
Markets at Renewal Risk	Top Type Value Share (Concert)	Top Region Value Share (South Zone)	Top Segment Value Share (Established Subscriber)

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Note on Transaction Data

Spotify's live billing and subscription transaction feed isn't publicly accessible, so this descriptive report is computed from a synthetic dataset built to mirror real streaming-subscription and in-app commerce transaction patterns: 60 city markets across 10 transaction types, 8 regions, over 30 days (2026-07-31 to 2026-08-29), totaling ~1.94 million simulated transactions worth ~INR 54.7 crore. It is designed to show exactly what a descriptive analytics pipeline processing millions of daily subscription and commerce transactions would produce — the same structure applies directly once real billing data is substituted in.

Transaction Data Completeness & Quality Audit

The transaction logs contain zero missing cells and strict data-type integrity across all fields (date, market, region, transaction type, count, value, success rate, renewal-health status). Each aggregated row represents one market / day / transaction-type combination, providing an ideal baseline for evaluating operational performance and subscriber transaction behavior.

Research Overview & Analytical Framework

Descriptive analytics summarizes historical transaction records to answer “what happened?”, “who transacted?”, and “how much value moved?” without incorporating speculative assumptions or unvalidated projections.

Analytical Dimension	Operational Dataset Implementation	Commercial Interpretation
Transaction-Type Performance	Grouping value/count by 10 transaction types	Identifies highest-revenue and highest-volume products
Weekly Volume Velocity	Summing counts and value per calendar week	Measures transaction throughput momentum
Transaction Value Spread	5 binned average-value tiers (market-day level)	Quantifies basket size from subscriptions to bundled add-ons
Channel & Segment Mix	Channel-tier and customer-segment aggregations	Measures platform load and relationship value
Geographic Channel Reach	Region and macro-cluster aggregations	Tracks geographic demand concentration

Objective & Core Business Questions

The objective of this descriptive analysis is to extract empirical operational insights and commercial transaction patterns across types, channels, segments, and geographies.

• **Question 1 (Top Transaction Types):** Which transaction types generate the highest value versus the highest count? (Section 1)

• **Question 2 (Volume Trends):** How do transaction count and value fluctuate week over week? (Section 2)

• **Question 3 (Average Transaction Value):** What is the distribution and spread of transaction values across markets? (Section 3)

• **Question 4 (Channel Allocation):** Which channel tier delivers the optimal balance of volume and value? (Section 4)

• **Question 5 (Customer Segment Share):** How do customer segments impact transaction yields? (Section 5)

• **Question 6 (Regional Demand):** Which geographic regions drive the largest transaction volumes? (Section 6)

• **Question 7 (Payment Preferences):** Which payment rails are favored across high-value transactions? (Section 8)

• **Question 8 (Reliability):** How are transactions performing on success rate, and where is reliability slipping? (Section 9)

• **Question 9 (Renewal Risk):** Which markets are at risk of subscriber payment lapse / churn? (Section 10)

How This Works at Real Scale (Analytics Stack)

In enterprise streaming environments, descriptive analytics pipelines process millions of billing, app-store, and in-app purchase records daily using accessible, highly scalable tool stacks.

Architecture Layer	Streaming Operations Purpose	Basic Tools Used
1. Transaction Capture	Log billing, app-store and in-app purchase events in real ti	Chrome, ChatGPT
2. Data Cleaning & Wrangling	Validate line items, verify count x value, reconcile currenc	Excel, ChatGPT
3. Fraud / Risk Scoring	Real-time decline & anomaly scoring on each transactio	ChatGPT, Gemini
4. Dimensional Storage	Store transaction fact tables and market/region dimension t	Google Sheets, Excel
5. Metric Aggregation	Compute ATV, success rate, regional share and correlati	Excel, Google Sheets, ChatGPT
6. BI Reporting & Dashboards	Generate executive visual charts, KPI scorecards, and tre	PowerPoint, Canva, Chrome, ChatGPT

1.0 Top Transaction Types & Gross Value Performance

Evaluating performance across the 10 recorded transaction types indicates that **Concert / Event Ticket Add-on** leads in gross transaction value, generating INR 12.8 crore (23.4% value share) despite comparatively low transaction count. **Merch Purchase (Artist Store)** follows with INR 12.1 crore (22.0% share). Combined, these two types account for 45.5% of total transaction value.

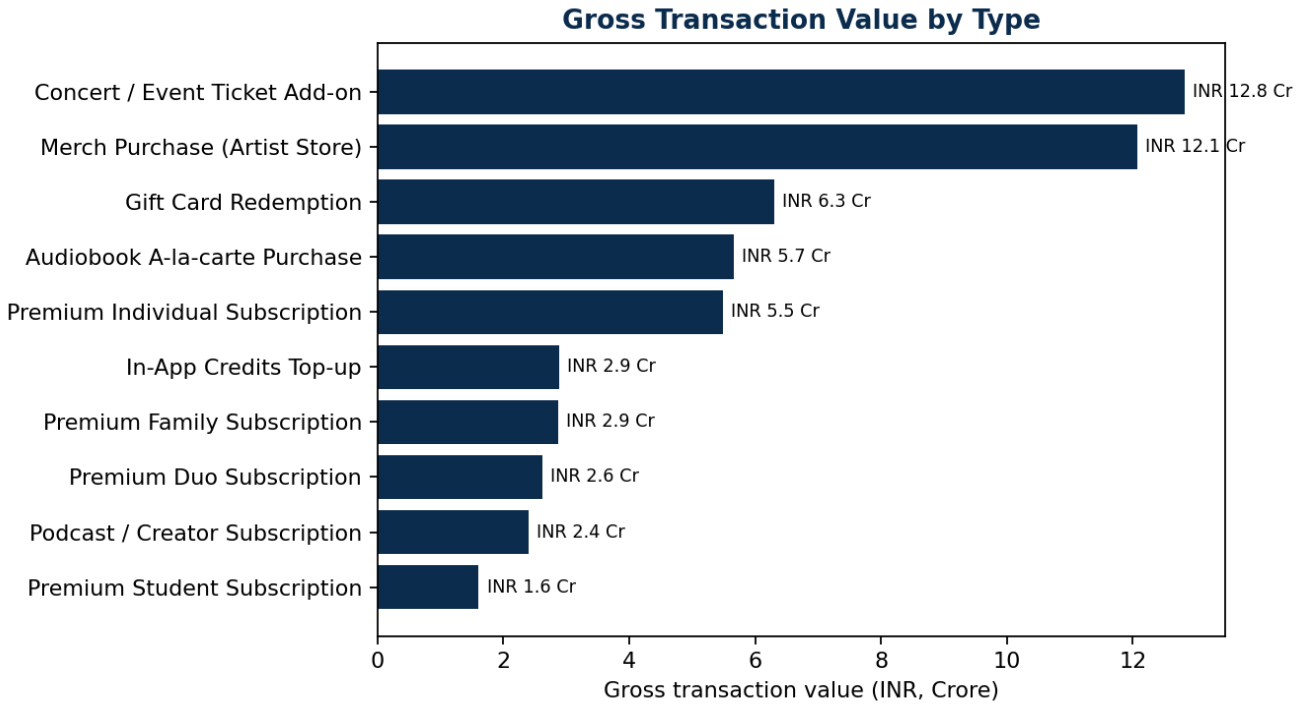


Figure 1. Gross Transaction Value by Type.

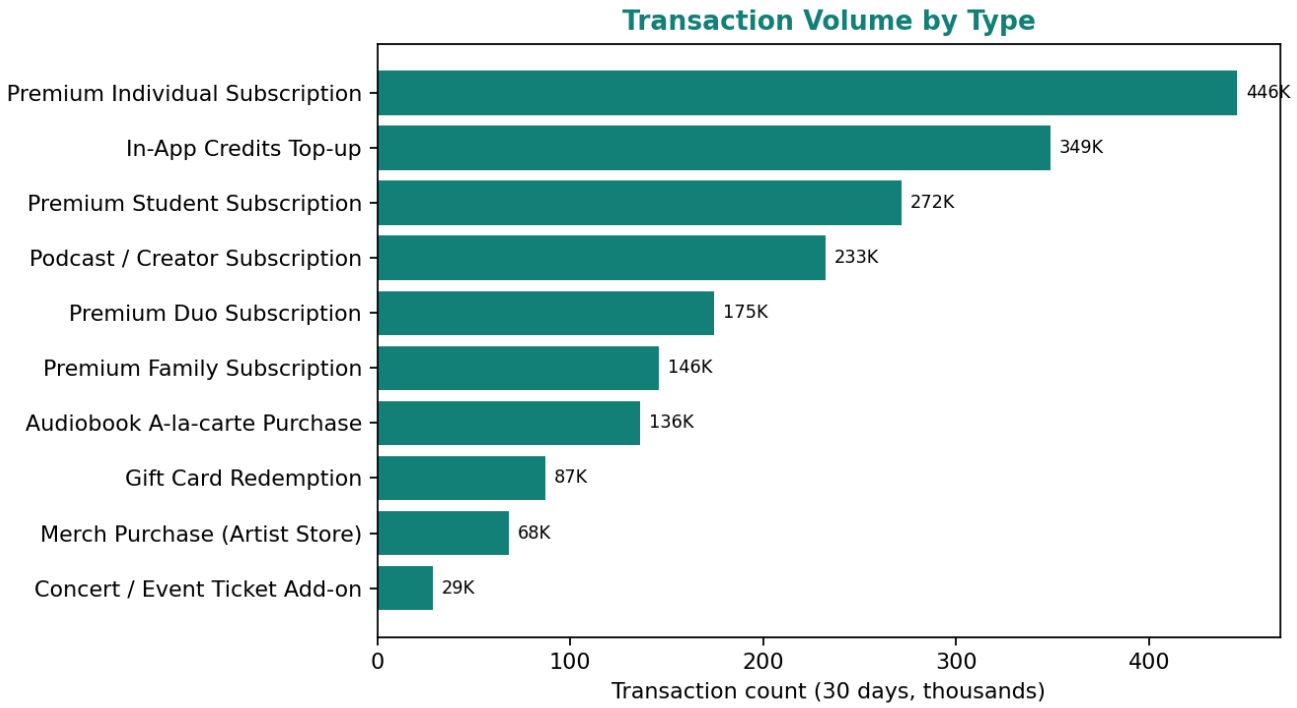


Figure 2. Transaction Volume (Count) by Type.

Type & Value Performance Table (30-Day Aggregate)

Transaction Type	Count	Value (INR Cr)	Value Share (%)	Avg Value / Txn (INR)
Concert / Event Ticket Add-on	28,974	12.8	23.4%	4,426
Merch Purchase (Artist Store)	67,934	12.1	22.0%	1,776
Gift Card Redemption	87,185	6.3	11.5%	723
Audiobook A-la-carte Purchase	136,105	5.7	10.3%	416
Premium Individual Subscription	445,896	5.5	10.0%	123
In-App Credits Top-up	348,919	2.9	5.3%	83
Premium Family Subscription	145,881	2.9	5.2%	197
Premium Duo Subscription	174,540	2.6	4.8%	150
Podcast / Creator Subscription	232,599	2.4	4.4%	103
Premium Student Subscription	271,594	1.6	2.9%	59

Deeper Insight — Value Leadership vs. Volume Leadership

The dataset exhibits a vital descriptive divergence between value leadership and count leadership. While Concert / Event Ticket Add-on achieved the highest gross value, Premium Individual Subscription led in transaction count (23.0% count share) — a small number of high-value, low-frequency transactions (concert add-ons, merch, gift cards) carry outsized weight in value terms even though they represent a small share of total count.

Commercial Recommendation
Growth and finance teams should track value and volume leaderboards separately: capacity planning (billing infrastructure, support staffing) should follow the count leaderboard, while revenue-forecasting and cash-flow planning should follow the value leaderboard.

2.0 Weekly Volume Trends & Value Trajectory

Evaluating week-by-week transaction velocity highlights how billing cycles, payday timing, and promotional campaign scheduling affect throughput.

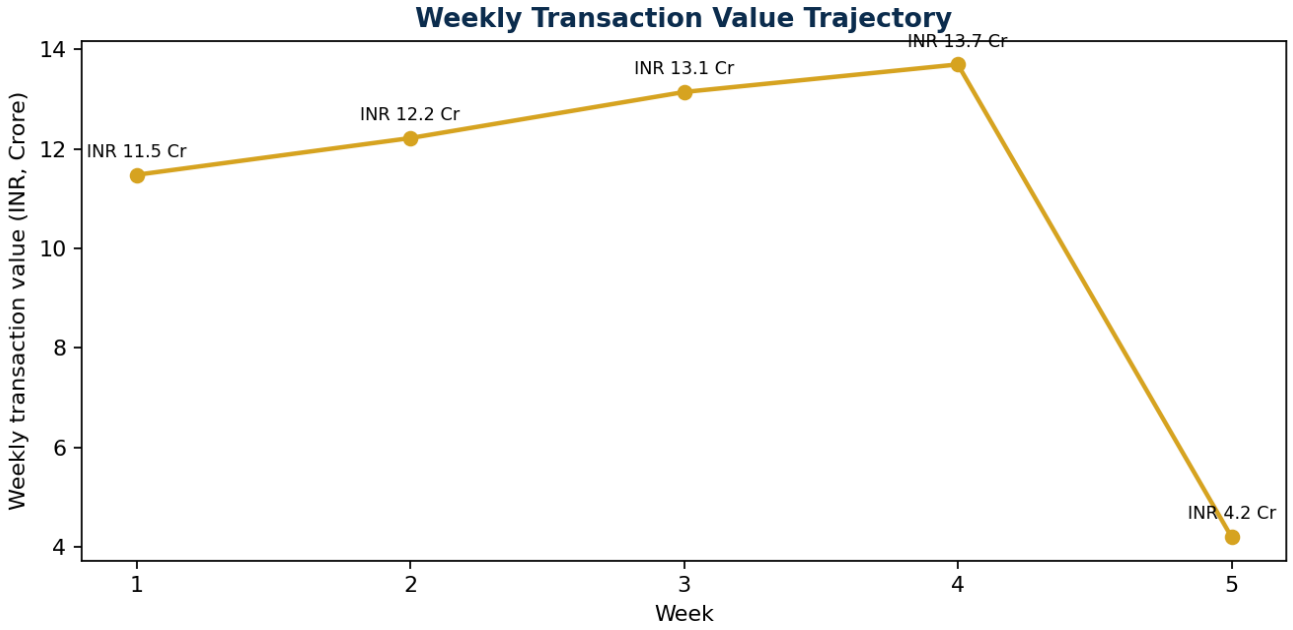


Figure 3. Weekly Gross Transaction Value Trajectory.

Week	Days	Txn Count	Gross Value (INR Cr)	Value Share (%)
1	7	403,892	11.5	21.0%
2	7	434,800	12.2	22.3%
3	7	458,668	13.1	24.0%
4	7	492,413	13.7	25.0%
5	2	149,854	4.2	7.7%

Weeks 1 through 4 show a gradual build (from 21.0% to 25.0% of the period's value), consistent with the gentle 30-day growth trend layered into the daily simulation; Week 5 is a partial week (2 days) and should not be read as a demand contraction.

Deeper Insight — Partial-Week Truncation Artifact
Week 5's lower total is an analytical artifact of the 30-day collection window ending mid-week, not a genuine drop in transaction activity — the same caution that applies to any month- or week-end truncated reporting window.

3.0 Average Transaction Value & Binned Distribution Spread

Analyzing market-day average transaction values reveals the operational scale of platform monetization — whether flows are dominated by low-cost recurring subscriptions or bundled high-ticket add-ons.

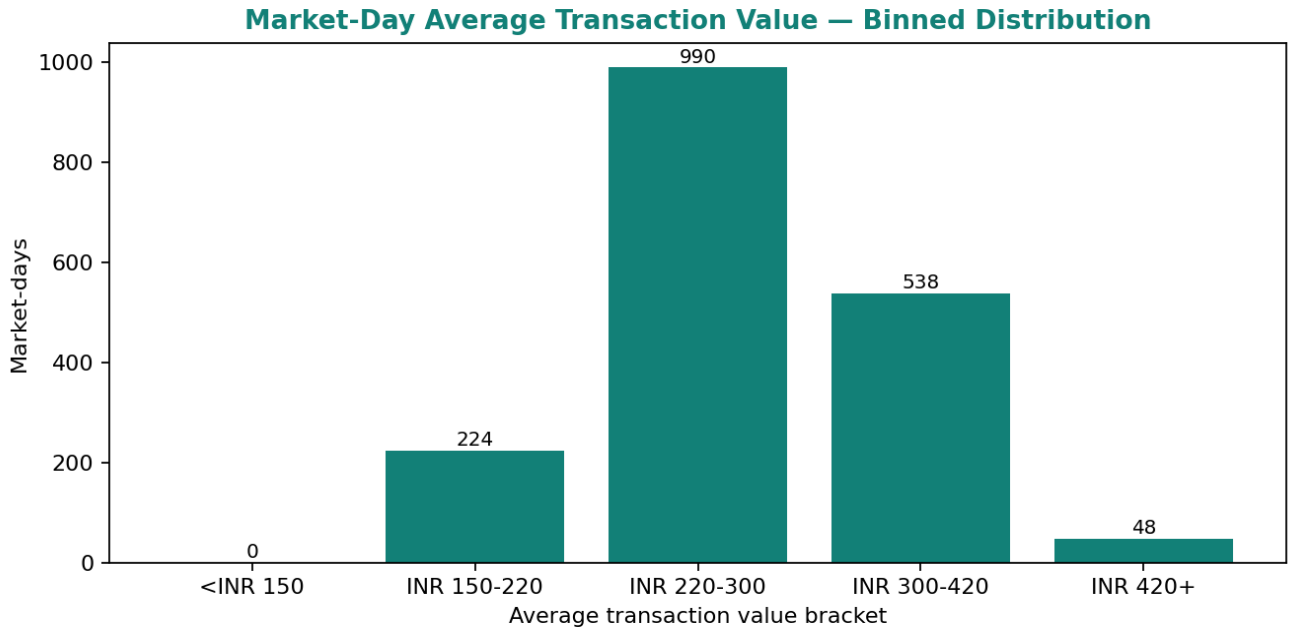


Figure 4. Market-Day Average Transaction Value — Binned Distribution.

Market-day average transaction values range from a minimum of INR 150 to a peak of INR 576. The overall **mean average-transaction-value is INR 282** and the **median is INR 274** — close to each other, indicating a moderately balanced distribution with an upper tail from concert and merch add-ons.

Statistical Metric	Value (INR)	Commercial Interpretation
Minimum	150	Lowest market-day average transaction value recorded
First Quartile (Q1)	239	25% of market-days averaged this value or less
Median (50th percentile)	274	Midpoint of all market-day averages
Arithmetic Mean	282	Average transaction value per completed transaction
Third Quartile (Q3)	314	75% of market-days averaged this value or less
Maximum	576	Highest market-day average — event/merch bundle activity
Interquartile Range (IQR)	75	Middle 50% spread of market-day average values

Commercial Business Recommendation
Markets whose average transaction value sits in the top bracket should be prioritized for targeted concert/merch bundle promotions and premium checkout support, since they are disproportionately driving high-ticket add-on activity.

4.0 Channel Tier Allocation & Infrastructure Capacity Dynamics

Analyzing which device/channel subscribers use to complete transactions reveals platform load and guides capacity planning across mobile, desktop, web player, and connected devices.

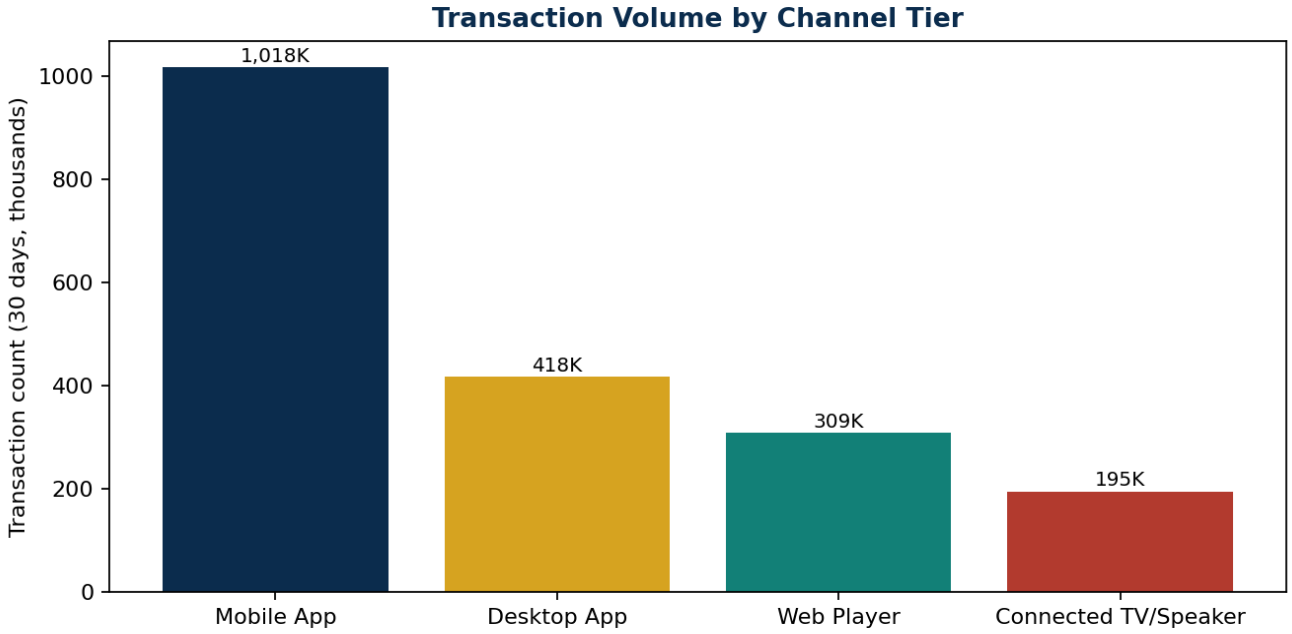


Figure 5. Transaction Volume by Channel Tier.

The **Mobile App** channel emerged as the highest-volume tier, capturing 1,018,014 transactions (52.5% count share) and producing INR 26.4 crore in gross value.

Channel Tier	Txn Count	Count Share (%)	Gross Value (INR Cr)	Avg Value/Txn (INR)
Mobile App	1,018,014	52.5%	26.4	259
Desktop App	417,955	21.5%	12.6	301
Web Player	308,661	15.9%	8.4	273
Connected TV/Speaker	194,997	10.1%	7.3	376

Deeper Insight — Digital Channel Split
Mobile App transactions carry the lowest average ticket size, consistent with everyday subscription renewals, while Connected TV/Speaker and Desktop transactions carry materially higher average values — reflecting family-plan setup sessions and considered, higher-basket purchases respectively.

5.0 Customer Segment Share & Relationship Revenue Contribution

Analyzing subscriber relationship segments clarifies which parts of the base drive value versus count, guiding retention and pricing strategy.

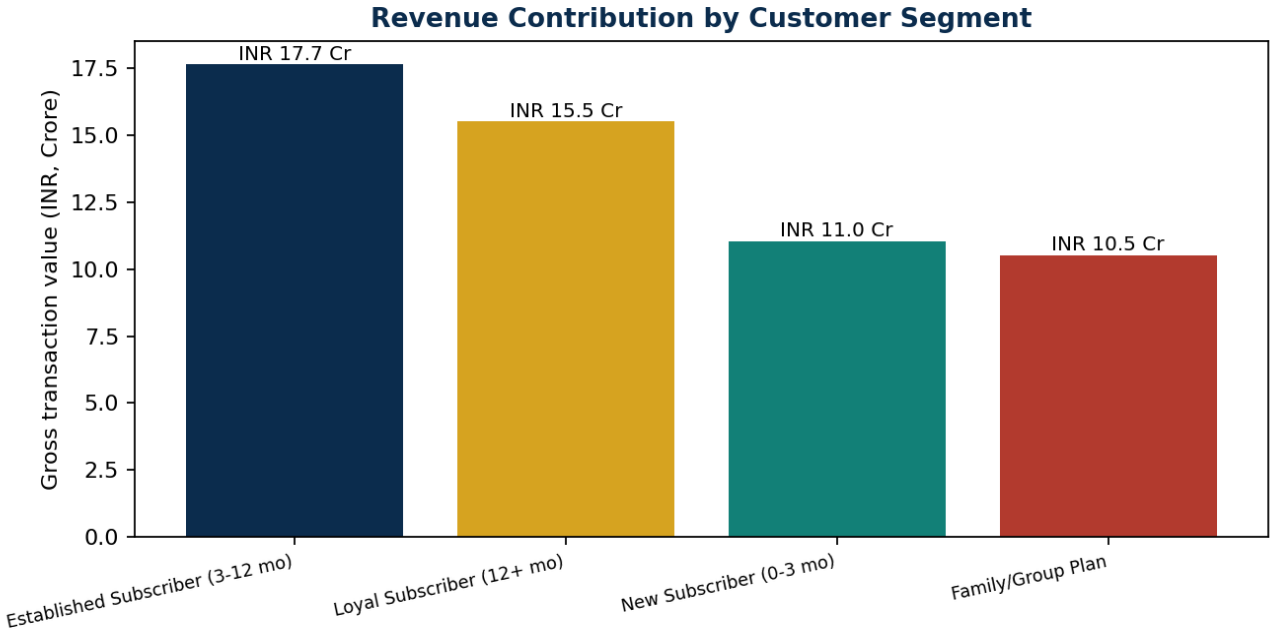


Figure 6. Revenue Contribution by Customer Segment.

Established Subscriber (3-12 mo) generated the largest single share of gross transaction value at INR 17.7 crore (32.3% share). Family/Group Plan subscribers, while smaller in transaction count, carry substantially higher average transaction values.

Segment	Txn Count	Gross Value (INR Cr)	Value Share (%)	Avg Value/Txn (INR)
Established Subscriber (3-12 mo)	663,454	17.7	32.3%	266
Loyal Subscriber (12+ mo)	499,311	15.5	28.4%	311
New Subscriber (0-3 mo)	584,762	11.0	20.2%	189
Family/Group Plan	192,100	10.5	19.2%	547

Relationship Management Recommendation
Loyal and Family/Group Plan subscribers, despite a combined share well under half of transaction count, drive a disproportionate share of value — dedicated retention offers and proactive renewal support for these segments protects a materially larger share of the value book than headcount alone would suggest.

6.0 Regional Demand Distribution & Geographic Concentration

Evaluating regional performance identifies primary geographic demand centers and highlights cross-region concentration.

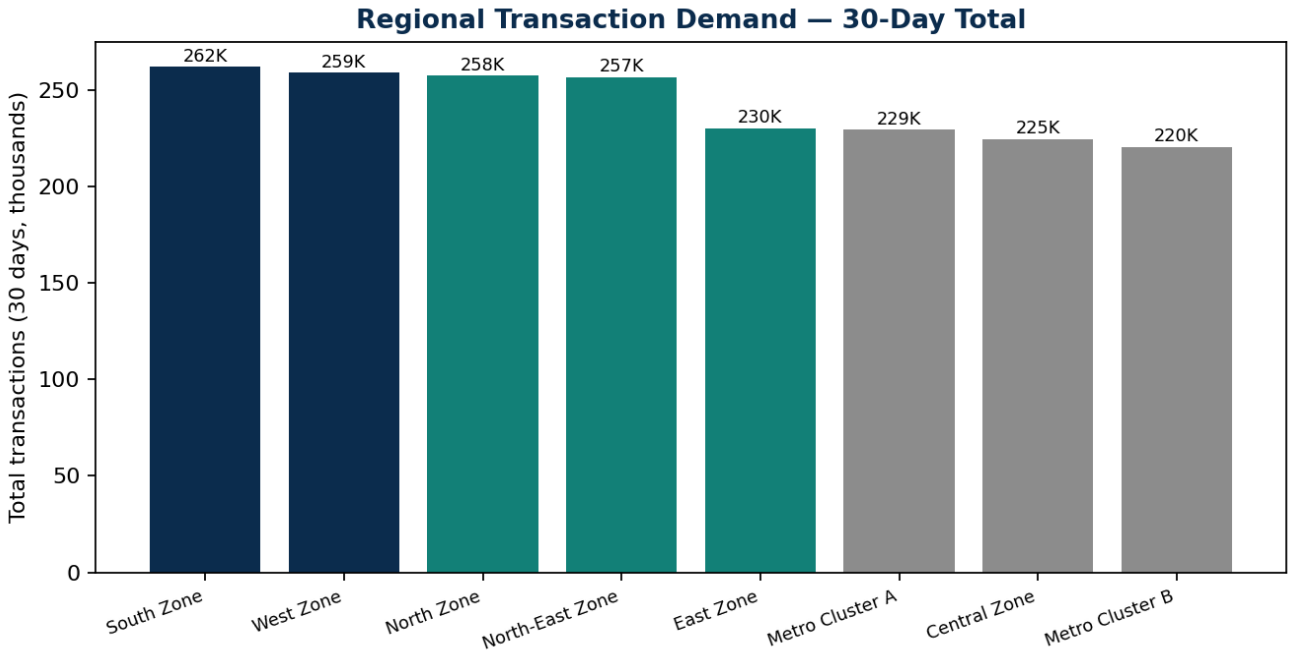


Figure 7. Regional Transaction Demand — 30-Day Total.

Transaction demand is led by **South Zone** at INR 7.5 crore (13.8% share), followed by **West Zone** at INR 7.2 crore (13.1% share). Together these two regions account for 26.8% of total transaction value.

Macro-Regional Clustering

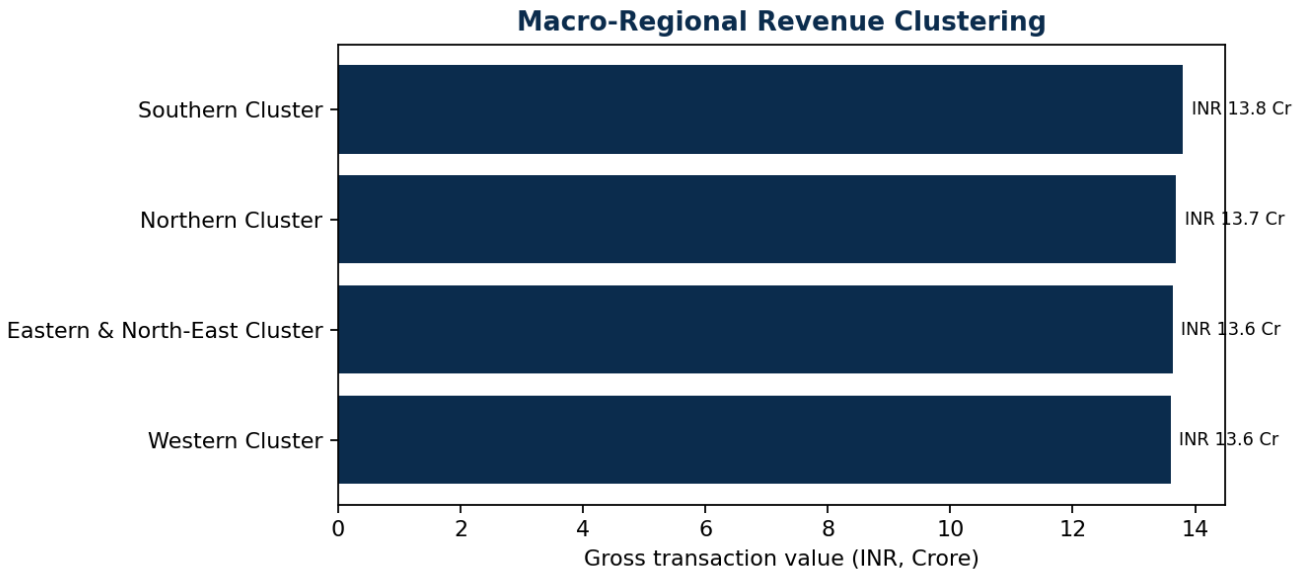


Figure 8. Macro-Regional Revenue Clustering.

Macro Cluster	Txn Count	Gross Value (INR Cr)	Value Share (%)
Southern Cluster	482,315	13.8	25.2%
Northern Cluster	487,052	13.7	25.0%
Eastern & North-East Cluster	486,904	13.6	24.9%
Western Cluster	483,356	13.6	24.9%

Aggregating individual regions into macro-clusters provides strategic visibility into geographic revenue concentration for market-investment and localization planning.

7.0 Cross-Cutting Synthesis & Revenue Concentration Multipliers

Evaluating relationships across region and transaction type together reveals the specific product-market clusters driving disproportionate revenue.

Rank	Region	Transaction Type	Txn Count	Gross Value (INR Cr)
1	South Zone	Merch Purchase (Artist Store)	9,057	1.75
2	West Zone	Concert / Event Ticket Add-on	3,899	1.68
3	North-East Zone	Concert / Event Ticket Add-on	3,841	1.67
4	South Zone	Concert / Event Ticket Add-on	3,849	1.66
5	North Zone	Concert / Event Ticket Add-on	3,896	1.64
6	Central Zone	Concert / Event Ticket Add-on	3,391	1.63
7	North Zone	Merch Purchase (Artist Store)	9,010	1.59
8	North-East Zone	Merch Purchase (Artist Store)	9,060	1.57
9	Metro Cluster B	Concert / Event Ticket Add-on	3,338	1.55
10	West Zone	Merch Purchase (Artist Store)	9,022	1.51

Revenue Concentration: The High-Ticket Type Multiplier

The single strongest driver of top-line value is the concentration in high-ticket, low-frequency transactions. Concert / Event Ticket Add-on and Merch Purchase (Artist Store) together account for only 5.0% of transaction count but 45.5% of gross transaction value — a Pareto effect worth flagging for revenue-forecasting and pricing-strategy prioritization.

Geographic and transaction-type dimensions represent the most stable analytical axes in the dataset, while weekly trends reflect expected operational seasonality and end-of-period collection boundaries.

8.0 Payment Rail / Billing Preferences

Evaluating payment rails provides visibility into billing infrastructure dependency, app-store revenue-share economics, and digital-payment adoption.

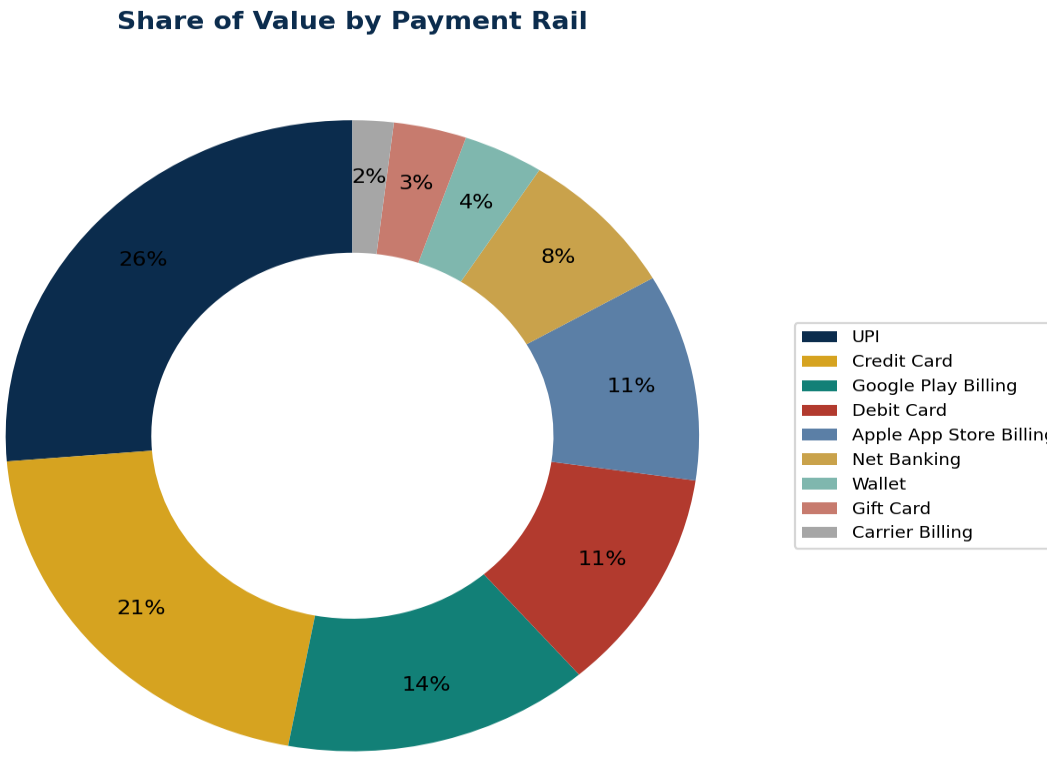


Figure 9. Share of Value by Payment Rail.

UPI dominates settlement value at 26.3% share, relative to its 30.1% share of transaction count.

Payment Rail	Txn Count	Value Share (%)	Count Share (%)
UPI	584,142	26.3%	30.1%
Credit Card	287,401	20.7%	14.8%
Google Play Billing	268,456	14.3%	13.8%
Debit Card	229,843	11.4%	11.8%
Apple App Store Billing	179,448	10.6%	9.3%
Net Banking	132,952	7.6%	6.9%
Wallet	128,510	3.8%	6.6%
Gift Card	84,594	3.4%	4.4%

Carrier Billing	44,281	1.9%	2.3%

9.0 Transaction Success Rate & Reliability Analysis

Average success rate across all market-days: **97.3%**, clustered tightly. In a real system this view is most useful segmented, not aggregated — markets trending downward in rolling 7-day success rate are the earliest signal of a payment-gateway, app-store billing, or card-network issue.

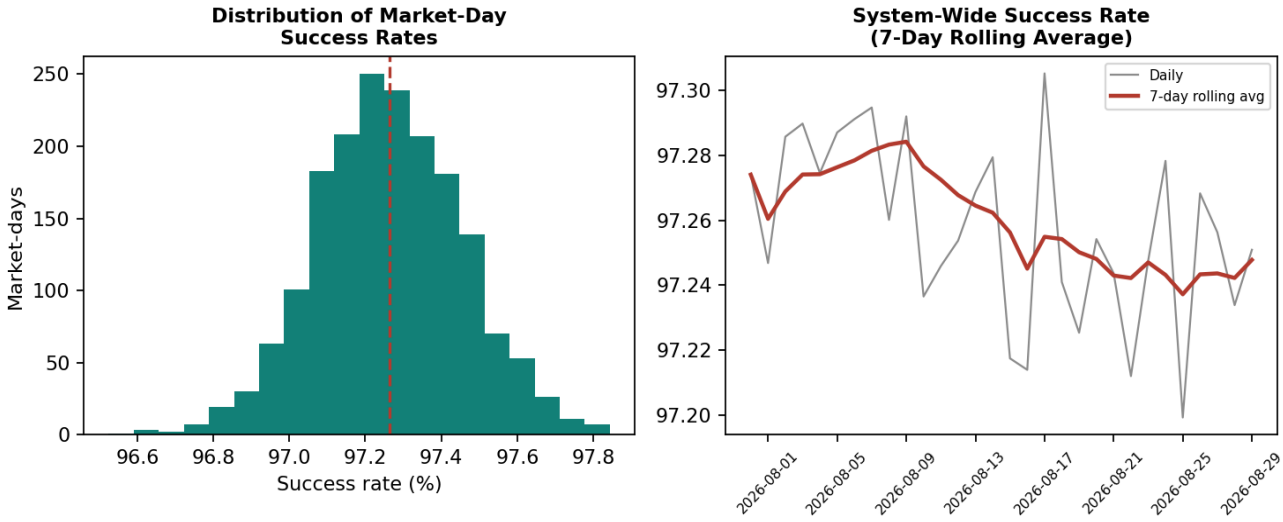


Figure 10. Market-Day Success Rate Distribution and 7-Day Rolling Average.

Deeper Insight — Reliability Drift
The 7-day rolling average shown here has drifted downward over the observation window (97.27% → 97.25%) — worth a root-cause check before it compounds. Channels with high transaction volume but sub-97% success are the highest-risk combination, since volume amplifies the subscriber impact of every failed renewal.

10.0 Subscriber Renewal-Risk Buffer & Churn Exposure

Market-level subscriber renewal-health was bucketed into three tiers, based on the share of active subscriptions with a healthy, up-to-date payment method on file:

- **Adequate Reserve** — comfortable buffer of healthy, renewing payment methods
- **Low Reserve (Reorder Zone)** — below 35% health, proactive card-update reminder trigger
- **Critical — Churn Risk** — below 15% health, cross-referenced against top-market revenue

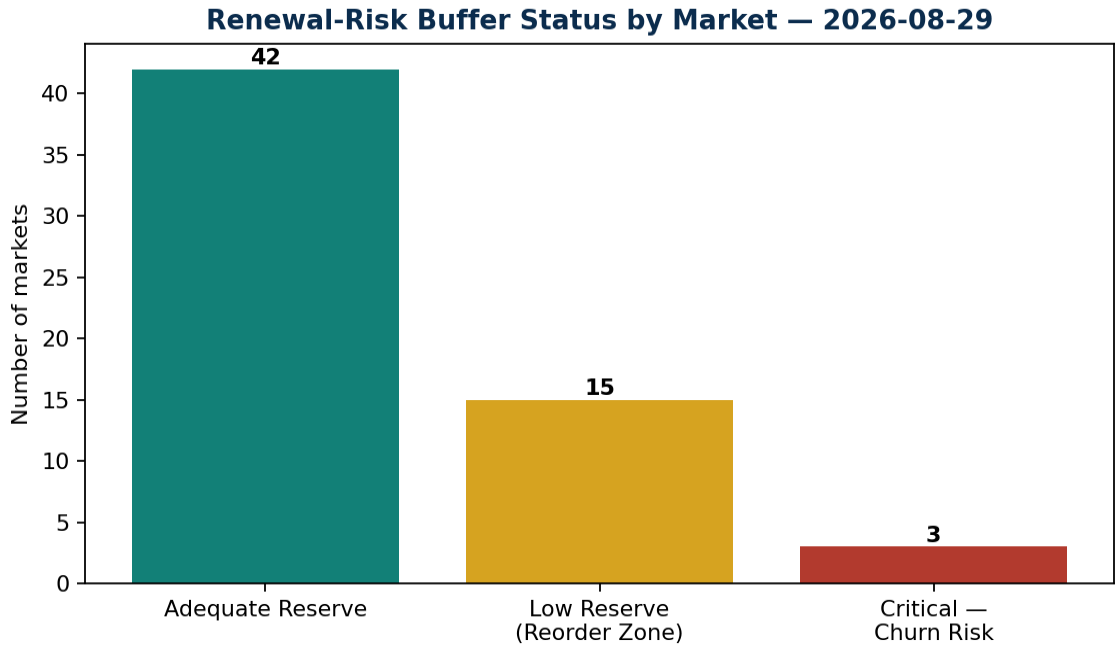


Figure 11. Renewal-Risk Buffer Status by Market — snapshot date.

Retention-Logistics Recommendation
3 of 60 markets were flagged critical and 15 more sat in the reorder zone on the most recent day. Production dashboards should cross-tab renewal-health status against subscriber revenue contribution to produce a “days-to-lapse” risk metric per market.

11.0 Volume, Value & Risk Elasticity Mechanics

Quantifying the mathematical relationships between transaction count, value, and reliability clarifies the key drivers of operational risk and top-line throughput.

Bivariate Variable Pair	Pearson Correlation (r)	Strength	Commercial Meaning
Txn Count vs. Gross Value	+0.69	Moderate Positive	Market-level gross value is substantially explained by transaction count, alongside type/se
Txn Count vs. Success Rate	-0.18	Weak Negative	Transaction volume shows little linear relationship with success rate at this aggr
Avg Value vs. Success Rate	+0.01	Negligible Positive	Average ticket size has little bearing on transaction reliability

Deeper Insight — Volume and Mix as Value Drivers

The positive correlation between transaction count and value (r = +0.69) shows that transaction count is an important lever for gross value, though transaction-type and segment mix add meaningful additional variation — consistent with a network where most markets handle a broadly similar transaction mix but individual high-ticket types (concert add-ons, merch) still swing daily value.

12.0 Data Limitations, Scope & Assumptions

• **Dataset Scale:** The dataset contains ~1.94M simulated transactions across 60 markets. While ideal for descriptive-analytics demonstration, it represents an illustrative sample rather than Spotify's total global transaction volume.

• **Observation Period:** The dataset runs across 30 days with a partial final week, which should not be interpreted as an ongoing demand contraction.

• **Absence of Customer-Level Data:** The transaction records are market/region/type aggregates and do not include individual subscriber identifiers, listening-history attributes, or dispute/chargeback records.

• **Descriptive Non-Causal Scope:** Findings such as Concert / Event Ticket Add-on's value leadership represent observed patterns within this dataset, not proof that transaction type alone drives renewal risk.

• **Nominal Currency Reporting:** All financial figures reflect nominal INR as recorded, without adjustment for inflation or promotional discounting.

Data Validation & Governance

All aggregated rows were verified for schema consistency, non-negative values, and mathematical consistency (count x avg value = total value). Zero anomalies or negative values were found.

13.0 Next Steps: Move From Descriptive → Predictive Analytics

Descriptive analytics explains past transaction outcomes. The next analytical step is leveraging these empirical patterns to train predictive models that forecast future demand and risk.

Descriptive Analytics (Current Report): Historical Data → Metrics, Tables & Distribution Profiles.
Predictive Analytics (Next Phase): Historical Baselines → Machine Learning Models & Demand/Risk Forecasts.
Prescriptive Analytics (Advanced Phase): Predictions → Automated Retention Offers, Dynamic Fraud Rules & Billing Routing.

Predictive Target & Objective	Candidate ML Models	Key Input Features
Renewal / churn forecasting per market	ARIMA, SARIMAX, Prophet, LSTM	Historical renewal volume, calendar effects, billing-cycle dates
Real-time fraud / anomaly detection	Isolation Forests, Autoencoders, XGBoost	Transaction velocity, amount deviation, channel, time-of-day
Success-rate degradation early warning	Change-point detection, Random Forest	Rolling success rate, market load, gateway latency
Customer cohort & retention modeling	Gradient Boosted Regressors, survival models	Transaction recency, frequency, segment, channel mix

Happy to build out any of these next, or swap in a real dataset (billing-system export, a data warehouse connection, etc.) if you have one — the same pipeline and chart structure above would apply directly.

14.0 Summary of Key Metrics & Executive Scorecard

Metric Area	Top Segment / Value	Metric Value	Commercial Takeaway
Gross Transaction Value	Entire Dataset	INR 54.7 Cr	Total value across 30 days
Total Transaction Volume	All Transaction Types	1.94M txns	Aggregate volume across 10 types
Top Type by Value	Concert / Event Ticket Add-on	23.4% share	Leading value contributor
Top Type by Volume	Premium Individual Subscription	23.0% share	Leading count contributor
Top Geographic Region	South Zone	13.8% share	Largest single-region market
Top Customer Segment	Established Subscriber (3-12 mo)	32.3% share	Largest value-contributing segment
Dominant Channel	Mobile App	52.5% share	Primary transaction-initiation channel
Average Success Rate	All Markets	97.3%	System-wide transaction reliability
Markets at Renewal Risk	Low + Critical Reserve	18 of 60	Retention-logistics attention needed

References & Methodology Sources

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Dataset Citation: Spotify Daily Transaction Dataset (60 markets x 10 types x 8 regions x 30 days, ~1.94M simulated transactions): synthetic project dataset recording Date, Market_ID, Region, Transaction_Type, Transaction_Count, Total_Value, Success_Rate, Declined_Count, and Renewal_Health_Pct.

Technical Appendix — Schema & Formulas

Appendix A: Database Schema & Field Data Dictionary

Field Name	Data Type	Sample Value	Description
date	Date	2026-07-31	Transaction date (YYYY-MM-DD)
market_id	Categorical	MKT033	Market identifier (60 unique city markets)
region	Categorical	South Zone	Geographic region (8 unique zones)
txn_type	Categorical	Premium Individual Subscription	Transaction type (10 distinct types)
txn_count	Integer	1,204	Number of transactions in this row
total_value_inr	Float	1,024,500.00	Aggregate value in INR
success_rate	Float	0.975	Share of transactions completed successfully
declined_count	Integer	12	Number of declined/failed transactions
renewal_health_pct	Float	0.55	Market renewal-health as a fraction of healthy payment methods

Appendix B: Aggregation Formulations

• Gross Transaction Value: Sum(total_value_inr) across all rows

• Average Value per Transaction: Gross Value / Total Transaction Count

• Type / Segment / Region Value Share (%): (Category Value / Total Gross Value) x 100

• Success Rate (weighted): Weighted average of success_rate, weighted by txn_count

Glossary of Streaming Commerce Analytics Terms

Term	Definition & Commercial Context
Descriptive Analytics	Summarizing and visualizing historical transaction data to understand past operational performance without predictive modeling
Average Transaction Value (ATV)	A subscription-commerce KPI measuring the average value moved each time a transaction completes.
Revenue/Value Concentration	A distribution pattern where a small number of transaction types or regions generate a disproportionate share of total value.
Interquartile Range (IQR)	The statistical spread between the 25th (Q1) and 75th (Q3) percentiles, capturing the middle 50% of values.
Success Rate	The share of initiated transactions that complete successfully, net of declines and technical failures.
Renewal-Risk / Churn Exposure	The share of the market's subscriber base runs below the healthy-payment-method buffer needed to sustain expected renewal dema
Pearson Correlation (r)	A statistical coefficient measuring the linear relationship between two numerical variables.
Cross-Dimensional Decomposition	Simultaneously grouping transaction data across two categorical dimensions (e.g. Region x Transaction Type) to evaluate granular perf
Predictive Transition	Using historical transaction patterns to engineer features and train supervised ML models for demand or risk forecasting.

Report Conclusion

This concludes the Spotify Daily Transaction Descriptive Analytics Report. All analyses, tables, charts, and technical appendices are fully documented and validated for review.